

Sonny Alberto Medina

Senior Data Scientist | Data Science Lead

Results-driven Senior Data Scientist with a PhD in Statistics and over 5 years of experience delivering high-impact machine learning and statistical models. Proven track record of spearheading complex data initiatives—such as driving \$80M USD in cost avoidance—and bridging the gap between advanced analytics and strategic business goals. Experienced in mentoring talent, guiding technical architectures, and partnering with cross-functional stakeholders. Eager to leverage expertise in quantitative risk, predictive modelling, and scalable AI to build and lead a high-performing data science team.

Professional Experience

Oct 2024 – **Sr. Data Scientist, Ford Motor Company, Remote**

- Present ○ **Technical Leadership & Business Impact:** Leading the development and end-to-end implementation of an advanced multi-signal forecasting system to predict vehicle warranty claims, driving up to \$80M USD in annual cost avoidance through early detection of quality defects.
- **Cross-functional Collaboration:** Partnering closely with engineering, quality, and executive stakeholders to align ML model outputs with strategic business interventions.
- **Advanced AI Architecture:** Architected speech-to-text pipelines and LLM-based semantic feature extraction to structure complex technician narratives, dramatically improving lead-time detection.
- **Innovation & Strategy:** Prototyping LLM-powered analytical agents to empower technical teams with root-cause exploration, laying the foundation for scalable AI-driven decision-making across the organization.

Sep 2021 – **PhD Candidate & Statistical Consultant, University of Bath, UK**

- Oct 2024 ○ **Consulting & Stakeholder Management:** Delivered actionable statistical insights through specialized consultancies with Roche, NU Mongolia, Schlumberger, and Wessex Water, translating complex mathematical findings into business value.
- **Research Excellence:** Authored two publications in top-tier probability journals, demonstrating deep expertise in complex stochastic modelling.
- **Mentorship & Communication:** Mentored undergraduate and master's students over 3 years of teaching Probability & Statistics, honing the ability to communicate highly technical concepts to varied audiences.

Nov 2018 – **Lead Data Scientist (Project), Nielsen, CDMX**

- Aug 2020 ○ **Project Management & Execution:** Led the technical redesign and algorithmic optimization of the Retail Establishments Survey for Mexico and Brazil, overseeing strategy from conception to deployment.
- **Process Optimization & ROI:** Successfully reduced operational costs by 7.5% in Mexico by automating and cutting a \$400K legacy workflow.
- **Performance Improvement:** Decreased overall survey cycle time by 3% while maintaining rigorous sampling accuracy and compliance.

CDMX, Mexico

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🌐 sdaphnis.github.io/sonny-presentation/ • **in** [sonny-medina-221451158](#)

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- Aug 2015 – **Data Analyst**, *Human Rights Commission*, CDMX
- Aug 2016
- **Automation & BI**: Spearheaded the automation of analytical reports and created performance-tracking dashboards using R Markdown.
 - **Operational Analytics**: Implemented complaint resolution metrics, leading data-driven initiatives to improve observer efficiency.

Education

- Sep 2020 – **Integrated PhD in Statistics**, *University of Bath*, UK
- Feb 2025 First class honours in the taught phase. Thesis: *Excursions from hyperplanes for α -stable processes*.
Supervisor: Prof. Andreas Kyprianou (University of Warwick) | [arXiv Link](#)
- Aug 2016 – **MSc in Probability & Statistics**, *CIMAT*, Guanajuato, MX
- Jun 2018 GPA: 9.50/10. Thesis: *Gerber-Shiu function for a renewal risk model with positive jumps...* | [Repo](#)
- Jul 2011 – **BSc in Actuarial Science**, *UNAM, Faculty of Sciences*, CDMX, MX
- Aug 2015 Graduated with honours. GPA: 9.0/10. Thesis: *INEGI 2010 Census sample: variance estimation*.

Leadership & Strategic Skills

- Management Cross-functional Collaboration, Technical Mentorship, Project Management, ROI Driven Delivery
- Financial/Risk Risk Quantification, Time Series Forecasting, Monte Carlo Simulation, Survey Sampling
- Data Science Custom ML Architectures, LLM Integration, NLP, Predictive Modelling, MLOps Awareness

Computer Skills

- Python Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, PySpark, SciPy
- R Tidyverse, Shiny, Survey, Plotly, Leaflet
- Data / Cloud SQL, GCP, Git, Advanced Analytics workflows

Awards & Publications

- 2024 *α -stable Lévy processes entering the half space or a slab*, Stochastic Processes and Applications
- 2024 *Gerber-Shiu Penalty Function for a Two-sided Renewal Risk Process*, Methodol Comp Appl Probab
- 2020 Nielsen Simply Excellence Award – *for leadership in saving \$30K USD and 60 workdays through modernizing survey algorithms and automated map generation*.

Languages

- Spanish Native
- English Fluent (CEFR C2) - Excellent business and academic proficiency

CDMX, Mexico